Project Management: Practice Problems on Network Diagram

MKT3717 Product and Brand Management

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Practice Problem 1.

Activity	Predecessor	Time (days)
A	--	8
B	--	6
C	--	3
D	A,B	10
E	C	8
F	A	5
G	D,E	3
H	G	4

1. Draw the network diagram. Find the critical path, and the expected completion time of this project.

Diagram omitted. Total 4 paths drawn (A-F, A-D-G-H, B-D-G-H, C-E-G-H).
Answer: A-D-G-H, 25 days

Practice Problem 1. (continued)

2. Disaster strikes and activity F takes 20 days instead of the anticipated 5 days. Using the table, how much longer will the project last than initially estimated?

Answer: 3 days (Now A-F becomes the critical path; with the length of 28 days)

3. Every day the construction crew on-site costs $1000. What activities would you consider crashing to reduce the project completion cost?

A) Activity E
B) Activity F
C) Activity G
D) All of these activities would be candidates for crashing.

Answer: C (Choose the activity on the critical path!)

4. How many days can activity C be delayed without changing the whole project's earliest completion time?

Answer: 7 days
(Slack = length of critical path (25) – length of the path where activity C is in (18))

Practice Problem 1. (continued)

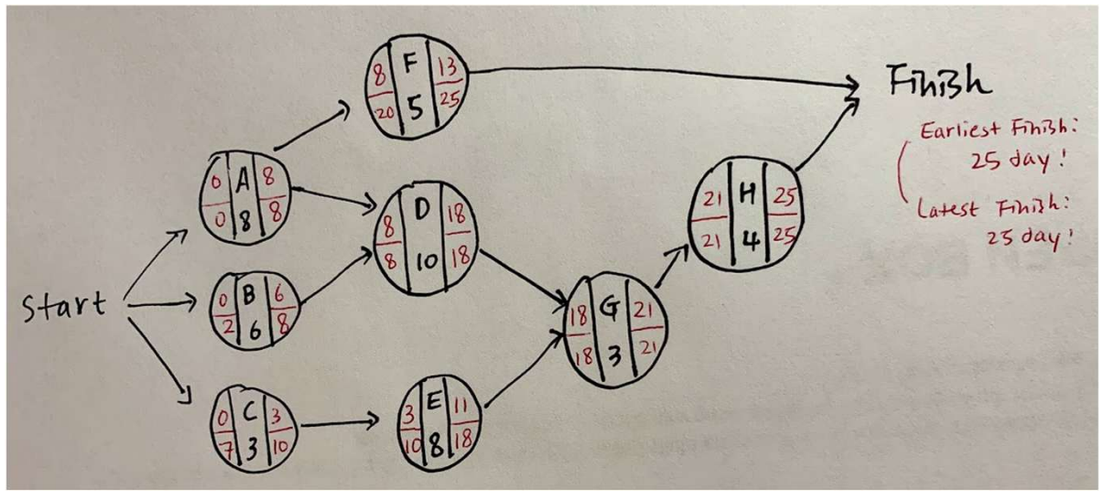
5. What is the latest start time for activity E?
Answer: Day 10

6. What is the earliest that activity D can be finished?
   A) Day 18
   B) Day 8
   C) Day 13
   D) Day 25

Answer: A. Refer to the next page network diagram and the scheduling chart.

Practice Problem 1. (continued)

Diagram:

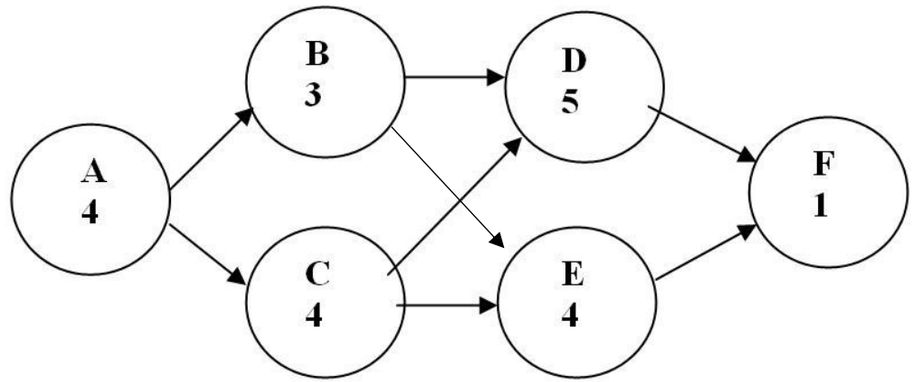


For Scheduling: You should be able to fill in the table below

Activity	Earliest Start	Earliest Finish	Latest Start	Latest Finish	Slack
A	0	8	0	8	0
B	0	6	2	8	2
C	0	3	7	10	7
D	8	18	8	18	0
E	3	11	10	18	7
F	8	13	20	25	12
G	18	21	18	21	0
H	21	25	21	25	0

Practice Problem 2.

- For the network diagram shown below, answer the following questions.



a. The critical path?

b. The expected length of the project?

c. Amount of slack time for each path?

Practice Problem 2. (continued)

- Length of each path
  - A-B-D-F: 4+3+5+1= 13
  - A-B-E-F: 4+3+4+1= 12
  - A-C-D-F: 4+4+5+1= 14
  - A-C-E-F: 4+4+4+1= 13

a. The critical path? A-C-D-F

b. The expected length of the project? 14 (days)

c. Amount of slack time for each path?

A-B-D-F: 1 day, A-B-E-F: 2 days, A-C-D-F: 0 day, A-C-E-F: 1 day

Glossary Summary

Activity:

- Project steps that consume resources and time
- The smallest unit of work effort consuming both time and resources that the project manager can schedule and control

Path:

- A sequence of activities that leads from the starting node to the finishing node (node – a connecting point)

Critical Path:

- The longest path; determines expected project duration

Critical Activity:

- Activities on the critical path

Activity Slack:

- The maximum length of time that an activity can be delayed without delaying the entire project